

END TERM EXAMINATION
THIRD SEMESTER [B.TECH] FEBRUARY 2023

Paper Code: CIC-211

Subject: Object Oriented Programming Using C++

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
1	Answer all the following questions briefly: (a) List the core features of Object-Oriented Programming. (b) Differentiate between a Constructor and Destructor in the context of class and object. (c) What are the various access specifiers in C++? How is the protected specifier useful? (d) What is generic programming? What are its primary advantages? (e) How does compile-time polymorphism differ from run-time polymorphism?	5 × 3 = 15
UNIT – I		
2	(a) Explain the meaning and syntax of an inline function. Write an inline function in C++. Further, write two situations when inline function expansion may not work. (8 Marks) (b) What is the difference between call by value and call by reference for a function? Explain with the help of C++ code. (7 Marks)	8 + 7 = 15
3	[OR] (a) Write any three features of a friend function and explain its functionality using a suitable example. Why should friend functions be avoided where possible? (7 Marks) (b) How does C++ support data abstraction and encapsulation? Appraise with an appropriate example why it is necessary to create good abstraction. (4 Marks) (c) Explain default parameter values in C++ with an example. (4 Marks)	7 + 4 + 4 = 15
UNIT – II		
4	(a) What do you mean by an array of objects? Explain how members of objects can be accessed in an array of objects with the help of a C++ program. (7 Marks) (b) Explain parameterized constructor and copy constructor with an example. (4 Marks) (c) Explain the state, identity and behaviour of an object in C++. (4 Marks)	7 + 4 + 4 = 15
5	[OR] (a) Explain the concept of operator overloading. Write a code in C++ to add two complex numbers using overloaded + operator. (7 Marks) (b) What is data hiding? What are the different mechanisms for protecting data from external users of a class's objects? (4 Marks) (c) Explain dynamic memory allocation in C++. (4 Marks)	7 + 4 + 4 = 15

[P.T.O.]

Q. No.	Question	Marks
	UNIT – III	
6	(a) Differentiate between overloaded functions and overridden functions in C++. What is ambiguity in inheritance? When do you encounter such a situation and how is it handled? Explain with an example. (8 Marks) (b) What do you mean by a template member function in C++? Write a program to define a function template for calculating the square of numbers with different data types. (7 Marks)	8 + 7 = 15
7	[OR] (a) Consider an example of declaring examination results. Design three classes: Student, Exam and Result. The Student class has data members such as roll number, name etc. Create the class Exam by inheriting the Student class. The Exam class adds data members representing the marks scored in six subjects. Derive Result from Exam class with its own data members such as total marks. Write an interactive C++ program to model this relationship. (8 Marks) (b) What are pure virtual functions? How do they differ from normal virtual functions? (7 Marks)	8 + 7 = 15
	UNIT – IV	
8	(a) Describe the various components of the Standard Template Library (STL) in detail. (8 Marks) (b) Write a C++ program to read the class object of student info such as name, age, gender from the keyboard and store them on a specified file using read() and write() functions. Again the same file is opened for reading and displaying the contents of the file on the screen. (7 Marks)	8 + 7 = 15
9	[OR] Write short notes on the following: (5 + 5 + 5 = 15 Marks) (a) Exception handling in C++ (5 Marks) (b) Generic Classes (5 Marks) (c) Vectors in STL (5 Marks)	3 × 5 = 15
